

Asad Ali Amjad

SOFTWARE ARCHITECT | EMBEDDED, BACKEND & AI/ML INFRA-STRUCTURE

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Results-driven Software Architect with 14 years of experience in embedded, backend services and AI/ML infrastructure development. Built a strong systems-level foundation in C/C++ and RTOS work, evolved through middleware and security backends into current focus on GPU-accelerated LLM serving platforms. Hands-on software architecture experience designing Provider and Factory abstractions, interface-based contracts, and event-driven systems. Recognized for delivering maintainable architectures, automating release workflows, and mentoring teams across embedded, backend, and ML infrastructure domains aligned with business goals.

Skills & Technical Proficiencies

Software Architecture: Provider pattern, Factory pattern, EventBus, interface-based design, multi-vendor abstractions, system-level integration, event-driven architecture

Programming & Scripting: C, C++, C#, Python, Java, PHP, SQL, JavaScript

AI/ML Infrastructure: NVIDIA NVML, vLLM, llama-cpp, HuggingFace, GGUF, multi-GPU inference, model lifecycle management

Embedded & Real-Time Systems: Nucleus RTOS, device drivers, multi-toolchain integration, hardware/software debugging (J-Link, ARM D-Stream, ST-Link, Mentor Embedded debug probes)

Graphics & UI Development: Qt integration, OpenGL/GLSL rendering, framebuffer & hardware-accelerated pipelines, UI automation/testing.

Backend & Enterprise Automation: C++, Python/Flask/FastAPI services, REST APIs, data aggregation, high-availability services, automated workflows.

Operating Systems & Platforms: Linux (Ubuntu, Oracle Linux), Windows, Nucleus RTOS

Networking & Middleware: OpenSSL, VPN, TCP/IP, Sockets, WebSockets, Socket.io, scrapy, multi-ACD/CX switch interfaces, enterprise middleware integration.

Databases & Messaging: MongoDB, SQL Server, Oracle, Redis, RabbitMQ

DevOps & Tools: Git/GitHub, Jenkins, CI/CD pipelines, Docker, GitHub Copilot, Eclipse IDE, VS Code, Google Test, Playwright, Automate App

Process & Leadership: Scrum/Agile, sprint planning, backlog management, mentoring engineers, cross-functional collaboration, customer engagement, technical documentation.

Professional Experience

ML System Software Engineer | think – Saudi Arabia

Mar 2026 – Present

- ✦ Designed and implemented the Provider pattern abstraction for GPU monitoring (NVML) and the Factory-based engine abstraction supporting multiple LLM backends (vLLM, llama.cpp), enabling vendor-extensible multi-GPU deployments.
- ✦ Built a model lifecycle management service: secure downloads from public model hubs, automatic GGUF/HuggingFace format detection, and centralized repository management.
- ✦ Delivered OpenAI-compatible API endpoints with auto-generated OpenAPI documentation, enabling external AI tool integration with the platform's LLM serving.
- ✦ Added KV cache pressure monitoring and llama.cpp request routing for multi-engine observability.
- ✦ Built thermal monitoring integration for liquid-cooled multi-GPU hosts to prevent overheating during sustained inference workloads.
- ✦ Built the web frontend (vanilla JS SPA) for GPU process management: model and engine controllers, offline mode with auto-reconnection, backend test simulator, and automated dependency installation.

Lead Software Engineer – Middleware & Switch Interfaces | Afiniti – Pakistan

Jun 2024 – Dec 2025

- ✦ Designed and implemented an **SHM size estimator** for Afiniti's proprietary in-house C/C++ shared memory library (user-space access via spin-locks, no syscall overhead after init), enabling capacity forecasting across dozens of **microservices** by deriving object-sizing formulas against a large internal API surface.
- ✦ Contributed to the **High Availability** team: maintained arbiter, watchdog, and data synchronization services,

resolved production issues, and implemented performance and reliability improvements for Afiniti's AI-driven call routing services.

- ✦ Owned switch interface services across multiple ACD/CX systems (primarily Nice In-Contact): maintained socket-based SDK, REST API, and WebSocket integrations; implemented **PII data protection**; and integrated **OpenTelemetry** metrics for SRE observability.
- ✦ Supported a customer migration from one ACD to another (Nice InContact), analyzing existing API implementations and documenting the integration in **OpenAPI format** to enable the customer's development team to implement middleware efficiently.

Principal Software Engineer – Cloud Backend | SlashNext – Lahore, Pakistan

Dec 2023 – May 2024

- ✦ Implemented features and bug fixes for cloud-hosted ML-powered phishing and malicious URL detection products serving enterprise email and browser extension users.
- ✦ Designed Python/Flask REST APIs and delivered backend features within the cloud backend team's established framework using JSON Schema validation, MongoDB, RabbitMQ, Redis, and Playwright in an **event-driven architecture**.
- ✦ Debugged and patched the existing Microsoft Playwright-based web scraping pipeline, and researched Chromium customization techniques to bypass Cloudflare bot-detection mechanisms blocking scraping workflows.

Senior Technical Lead Engineer - Middleware | Mentor Graphics – Lahore, Pakistan

Feb 2021 – Nov 2023

- ✦ Served as **Scrum Master** for embedded and middleware projects, optimising sprint planning, backlog grooming, stand-ups, and retrospectives. Authored article on [Sprint Planning Framework — True Sprint Velocity](#).
- ✦ **OpenSSL upgrade automation**: Streamlined OpenSSL upgrade integration for Nucleus RTOS by automating 90% of upgrade steps and 70% of testing, cutting release cycles from months to a single 3-week sprint.
- ✦ Developed automated Jenkins pipelines to execute OpenSSL unit tests on Nucleus across QEMU and multiple ARM-based SoC hardware platforms, ensuring consistent build quality.
- ✦ Managed **CVE fixes** security patches across parallel Nucleus versions, interpreting RFCs, implementing fixes, and integrating into the **Scaptic framework** for uniform functional testing on multiple embedded hardware platforms, strengthening system integrity for legacy customers.

Technical Lead Engineer – Graphics & Middleware | Mentor Graphics – Lahore, Pakistan

Aug 2016 – Jan 2021

- ✦ Collaborated in **cross-functional R&D projects**, including graphics testing automation and custom demos.
- ✦ Engineered the porting of OpenSSL 1.1.1 to Nucleus 3.x and 4.x, enhancing system security and compatibility.
- ✦ Ported a FAT RAM Disk driver to Nucleus 4.1, mounting into Nucleus 4.1 using parameters from Device Tree Blob (DTB) for the underlying SoC.
- ✦ Owned and enhanced the Python-based **Scaptic framework** for functional testing of Nucleus networking stack; designed and implemented automated tests for new VPN features using Android clients, **adb**, and **Automate App** integration.
- ✦ Developed a streamlined, repeatable process for **Qt open-source upgrades integration into Nucleus RTOS**, slashing manual completion time from months to weeks.
- ✦ Fixed bugs in an **Eclipse plugin using Java**, contributing to improved IDE stability.

Senior Software Development Engineer – Graphics | Mentor Graphics – Lahore, Pakistan

Oct 2014 – Jul 2016

Award: Best Technical Support for outstanding technical delivery.

- ✦ Integrated **Qt open-source with Nucleus RTOS**, providing framebuffer-based and **OpenGL**-based rendering options while handling POSIX and non-POSIX configurations for maximum compatibility.
- ✦ Devised high-fidelity **OpenGL/GLSL demos**, enabling performance benchmarking of Qt with GPU on Nucleus RTOS.
- ✦ Optimised Qt C++ App performance on Nucleus RTOS, achieving measurable improvements in efficiency, reliability, and user experience for medical devices.
- ✦ Developed a fully functional Qt C++ App to demo industrial tank automation.

Software Developer – Control Systems / Robotics | AEDesign – Lahore, Pakistan

Aug 2013 – Oct 2014

Product: AVCASS Mini Robot / TS-7800 Embedded Linux C++ Application / Windows C# Application

- ✦ Developed a C++ application with integrated PID control for robotic linear motion.
- ✦ Engineered network communication modules enabling real-time robot control via TCP/IP and UDP/IP.
- ✦ Delivered a Windows hybrid control application in C# for rapid communication and autonomous navigation.

- ✦ **SQL query performance:** Automated a frequently requested report by designing **SQL Server aggregation tables** integrated into nightly sync processes, reducing a 15-minute 300-line query to a 30-second 15-line query, using C# and ASP.NET.
- ✦ Collaborated with local management teams to translate reporting requirements into **optimized data structures** that improved query performance and data accessibility.

- ✦ Delivered a ready-to-use shopping-cart web part for e-commerce enablement in Microsoft SharePoint sites, using C# and ASP.NET. Used CAML and LINQ.

*Additional Experience as **Graduate Teaching Assistant (Aug 2011 – Jan 2012) at LUMS – Lahore, Pakistan***

Additional Early Experience / Projects

- *Developed web applications in PHP using CodeIgniter framework during undergraduate studies.*
- *Completed extensive academic projects in Java, including object-oriented design and multi-threaded applications.*
- *Completed extensive academic projects in embedded C on different MCU platforms like Atmel, Microchip, etc. and Verilog projects on FPGA kits.*

Education

Bachelor of Science (Honours) in Computer Engineering | LUMS – Lahore, Pakistan

Jun 2011